

CHEMICAL KINETICS

Rate of Reaction

“The rate of chemical reaction is the change in the concentration of reactants and product per unit time”

$$\text{Rate of Reaction} = \frac{\text{Increase in the product concentration}}{\text{Unit time}}$$

$$\text{Rate of Reaction} = \frac{\text{Decrease in the reactant concentration}}{\text{Unit time}}$$

If dx is very small change in concentration of a product in a very small-time interval dt , the rate of reaction is expressed as

$$\text{Rate of reaction} = dx / dt$$

Consider a general reaction



The rate of reaction can be expressed in term of the rate of disappearance of reactant A or the rate of appearance of product B.

$$dx/dt = -d[A] dt$$

$$dx/dt = + d[B] dt$$

Where $d[A]$ and $d[B]$ represent changes in the concentration of A and B respectively. The negative sign indicates a decrease in the concentration of the reactant A, whereas the positive sign indicates the increase in the concentration of product B.

Rate Law

“Rate of reaction is defined as the instantaneous change in concentration of a reactant or product at a given time.”

Or

“The rate of chemical reaction is proportional to the molar concentration of reactants each raised to a power.”

Thus for a general reaction:



$$\text{Rate} \propto [A]^x$$

$$\text{Rate} = k [A]^x \quad \text{Rate law or rate equation}$$

Where k is proportionality constant and is known as rate constant.

The exponent x in the rate equation is called order of reaction with respect to reactant A.

Rate constant

“Rate constant can be defined as the rate of reaction when the molar concentration of each reactant is unity.”

Each reaction has its own specific rate constant, independent of concentration and time.

But the value of the rate constant changes with temperature.

Rate Law or rate equation:

"An expression which shows how the reaction rate is related to the concentration of reactants is called the rate law or rate equation. "

$$\text{Rate} = k [A]^x$$

Order of Reaction and Rate Equation

“Order of reaction may be defined as the number of molecules of reactants participating in the rate determining-step.”

"Order of reaction may be defined as the sum of all the exponents to which the molar concentration terms in the rate equation are raised".

Explanation:

Consider a general reaction between a moles of A and b moles of B to give c moles C and d moles of D.



The rate equation can be written as

$$\text{Rate} \propto [A]^a[B]^b$$

$$\text{Rate} = k [A]^a[B]^b$$

The exponent 'a' is the order of reaction with respect to species 'A' and the exponent 'b' is the order of reaction with respect to species 'B'.

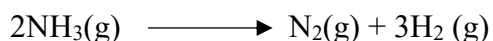
The sum 'x+y' is called the overall order of the reaction.

Types of Order of Reaction**a) Zero order reaction**

A reaction that is independent of the concentration of reactant molecules is called zero order reaction.

Example

1. An example is the decomposition of ammonia on heated tungsten.



$$\text{Rate} = k[\text{NH}_3]^0$$

The concentration of ammonia decreases continuously until it reaches zero.

2. The combination of H_2 and Cl_2 in presence of sunlight is also a zero-order reaction.

3. Reactions catalyzed by enzymes also follow zero-order kinetics.

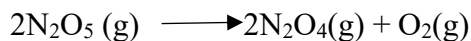
B) First order reactions:

“A reaction whose rate of reaction is directly proportional to the first power of the concentration of single reactant molecule is called the first order reaction.”

$$\text{Rate} \propto [\text{A}]^1$$

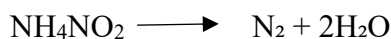
Examples:

i) Thermal decomposition of N₂O



$$\text{Rate} = k [\text{N}_2\text{O}]$$

ii) Decomposition of ammonium nitrite in aqueous solution.



$$\text{Rate} = k[\text{NH}_4\text{NO}_2]$$

C) Second order reactions

“A reaction for which sum of exponents of rate equation is two is called the second order reaction.”

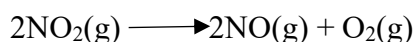
Equation:

$$\text{Rate} \propto [\text{A}]^2$$

$$\text{Rate} \propto [\text{A}]^1 [\text{B}]^1$$

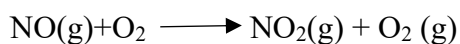
Examples:

i) Decomposition of nitrogen dioxide.



$$\text{Rate} = k [\text{NO}_2]^2$$

ii) Oxidation of NO



$$\text{Rate} = k [\text{NO}] [\text{O}_3]$$

$$\text{Order of reaction} = 1 + 1 = 2$$

iii) Reaction of NO₂ with ozone:



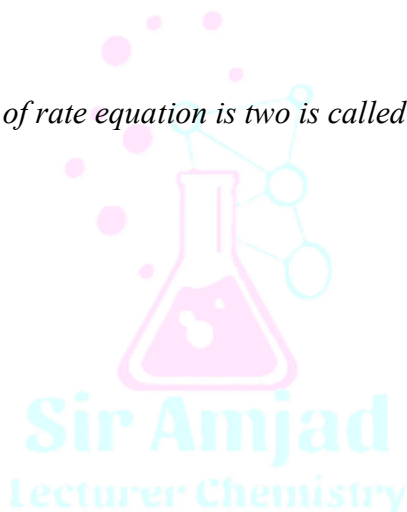
$$\text{Rate} = k [\text{NO}_2] [\text{O}_3]$$

D) Third order reactions

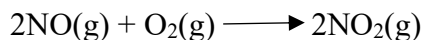
“A reaction for which sum of exponents of rate equation is three is called the third order reaction.”

$$\text{Rate} \propto [\text{A}]^3 \text{ or}$$

$$\text{Rate} \propto [\text{A}]^2 [\text{B}]^1 \text{ or}$$



$$\text{Rate} \propto [\text{A}]^1 [\text{B}]^1 [\text{C}]^1$$

Examples**i) The oxidation of NO by O₂**

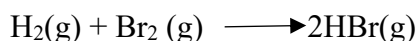
$$\text{Rate} = k [\text{NO}]^2 [\text{O}_2]$$

ii) $2\text{FeCl}_3(\text{aq}) + 6\text{KI}(\text{aq}) \longrightarrow 2\text{FeI}_2(\text{aq}) + 6\text{KCl}(\text{aq}) + \text{I}_2(\text{s})$

$$\text{Rate} = k [\text{FeCl}_3][\text{KI}]^2$$

F) Fractional order reactions:

A reaction for which the sum of exponents of rate equation is in fraction is called the fractional order reaction.

i) The reaction between H₂ and Br₂ to produce HBr is a half order in Br₂ and first order in H₂.

$$\text{Rate} = k [\text{H}_2] [\text{Br}_2]^{1/2}$$

Order of reaction is $1 + 0.5 = 1.5$

ii) $\text{CHCl}_3(\text{l}) + \text{Cl}_2(\text{g}) \longrightarrow \text{CCl}_4(\text{l}) + \text{HCl}(\text{g})$

$$\text{Rate} = k [\text{CHCl}_3] [\text{Cl}_2]^{1/2}$$

G) Pseudo First order reaction

A bimolecular reaction for which solvent is in excess and its concentration remains constant and does not take part in rate determining step is called a Pseudo First order reaction.



$$\text{Rate} = k[(\text{CH}_3)_3\text{C-Br}]$$

Determination of Order of Reaction, rate law and rate constant

The effect of a change in reactant concentration on reaction rates cannot be derived from a chemical equation. It can only be determined experimentally by determining the order of chemical reactions.

Example:

Jet engines release Nitrogen (II) oxide in the upper atmosphere. In the ozone layer of upper atmosphere Nitrogen (II) oxide reacts with ozone to form nitrogen (IV) oxide and oxygen.



The following data was obtained for this reaction at 25 °C.

Experiment (moles dm ⁻³ s ⁻¹)	Initial [NO]	Initial [O ₃]	Initial rate
1	1.00×10 ⁻⁶	9.00×10 ⁻⁶	1.98×10 ⁻⁴

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2	2.00×10^{-6}	9.00×10^{-6}	3.96×10^{-4}
3	1.00×10^{-6}	3.00×10^{-6}	6.60×10^{-5}

Use this data to determine the rate law for the reaction.

Rate of reaction with respect to [NO]:

Initial concentration of [NO] double in experiment 1 and 2 but initial concentration of O₃ kept constant.

Experiment 1 Experiment 2

1.98×10^{-4} 3.96×10^{-4}

$$\frac{1.98 \times 10^{-4}}{1.98 \times 10^{-4}} = \frac{3.96 \times 10^{-4}}{1.98 \times 10^{-4}}$$

1:2

Rate $\propto$ [NO]

In experiment 1 and 3 the initial concentration of [NO] kept constant but initial concentration of [O₃] become one third.

$$\frac{1.98 \times 10^{-4}}{1.98 \times 10^{-4}} = \frac{6.6 \times 10^{-5}}{1.98 \times 10^{-4}}$$

1:1/3

Rate $\propto$ [O₃]

Rate law

Rate $\propto$ [NO][O₃]

Rate = k [NO][O₃]

By using data from experiment 1

$$1.98 \times 10^{-4} = k[1 \times 10^{-6}][9 \times 10^{-6}]$$

$$1.98 \times 10^{-4} = k[9 \times 10^{-12}]$$

$$K = \frac{1.98 \times 10^{-4}}{9 \times 10^{-12}}$$

$$K = 2.2 \times 10^7 \text{ mol.s.dm}^{-3}$$

Half-life method for determining the order of reaction:

The half-life ($t_{1/2}$) of a reaction is the time takes for the concentration of a reactant is reduced to half.

The relationship between half-life and rate constant depends on the order of reaction

When

$$n = 0 \quad t_{1/2} = [A_0]/2k.$$

Where [A₀] = initial concentration of reactant

$$n = 1 \quad t_{1/2} = 0.693/k$$

$$n = 2 \quad t_{1/2} = 1/k [A_0]$$



Consider the following reaction that follows 1st order kinetic, its half-life is 30s $A + 2B \rightarrow C$

Rate constant for this reaction can be determined as follows.

$$t_{1/2} = 0.693/k$$

$$30 = 0.693/k$$

$$K = 30/0.693$$

$$k = 0.0231 \text{ s}^{-1}$$

Therefore, the rate constant for the reaction is, $k = 0.0231 \text{ s}^{-1}$

Effect of temperature on the rate of reaction

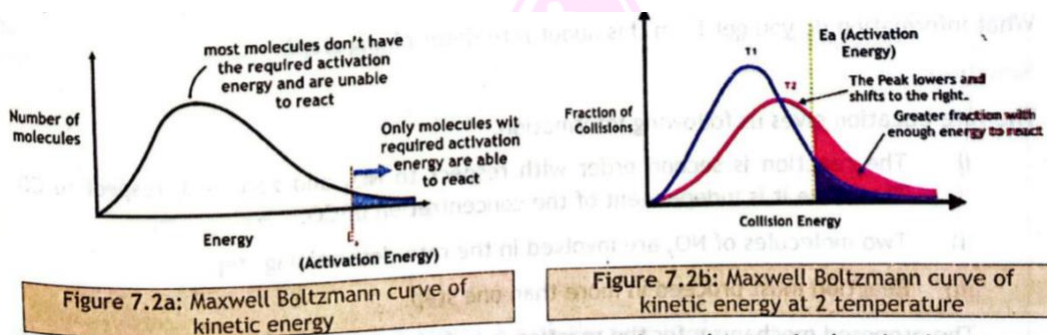
The reaction rate generally increases with increasing temperature. According to collision theory, the rate of a reaction is proportional to the number of collisions between the reacting molecules.

An increase in temperature increases the average kinetic energy of the molecules. This increases the average speed of the reacting molecules. An increase in the kinetic energy of the reacting molecules increases the collision frequency, i.e., the number of effective collisions and thus the reaction speeds up.

But only effective collisions produce a reaction.

For the collision to be effective, the molecules must have activation energy and must be correctly oriented.

The number of molecules with kinetic energy at least equal to E_a at temperature T is proportional to the shaded area under the Maxwell Boltzmann kinetic energy curve.



The distribution curve shows that most molecules do not have the required activation energy at T_1 or lower temperature. Only some of the molecules that required the activation energy are able to react.

When the temperature increases (from T_1 to T_2), the energy of the molecules also increases. Thus, the proportion of molecules with the required activation energy increases.

It was found that, in general, the reaction rate increases two to three times for every 10 K increase in temperature.

Arrhenius Equation:

Arrhenius (1889) studied the effect of temperature on reaction rates. He found that the effect of temperature on rate of reaction is given by the following equation.

$$k = Ae^{-E_a/RT}$$

k = rate constant,

E_a is energy of activation,

R is gas constant ($R=8.3143 \text{ JK}^{-1} \text{ mole}^{-1}$)

A is constant known as Arrhenius constant.

Rate constant k varies with the temperature. It increases with temperature which in turn increases rate of reaction.

Mechanism of chemical Reaction

The path that reactants take to form products in a chemical reaction is called the mechanism.

If a reaction proceeds in two or more steps, one of the steps is the slowest. The rate of the slowest step determines the overall reaction rate.

Rate determination step

"The slowest step in the reaction mechanism, which determines the overall rate of the reaction, is called the rate-determining step."

Example 7.4:



$$\text{Rate} = k[\text{NO}_2]^2$$

Solution.

The proposed mechanism for the reaction is as follows.

Step I



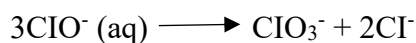
Step II



The first step is the rate determining step. Species NO, that does not appear in the overall reaction is called give any intermediate.

Example 7.5:

Hypochlorite ion ClO^- in aqueous solution decomposes to chlorate ion ClO_3^- and chloride ion.



The rate of the reaction is second order in ClO^- ion

$$\text{Rate} = k [\text{ClO}^-]^2$$

The following two-step mechanism is consistent with the rate law for the reaction.





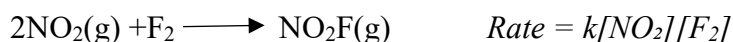
Overall reaction



Select the rate-determining step.

Potential Energy Diagram and Reaction Mechanism

The experimental rate law for this reaction is given below:



Step 1: Determine the number of elementary steps.

Step 2: Determine the rate-setting step.

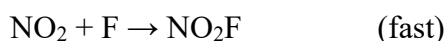
Because the activation energy of step 1 is greater than the activation energy of **step 2**: Therefore, step 1 is a slow and rate-determining step.

Step 3: Use the rate law to determine the number of molecules involved in the rate-determining steps

Step 4: Propose the two main steps of the mechanism.

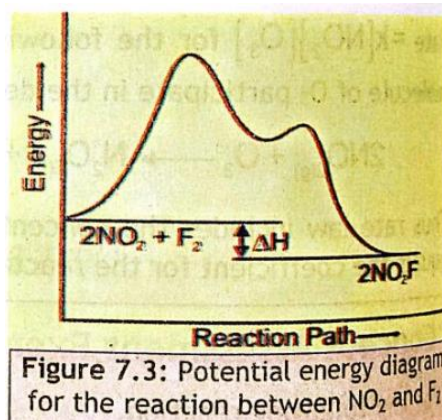
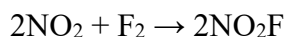
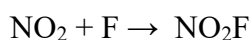
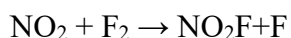
Reaction intermediate

"A species that is produced in one step of a chemical reaction and consumed in another step is called a reaction intermediate."



Step 5:

Add the two steps to get the overall reaction which must be same as the reaction under consideration.

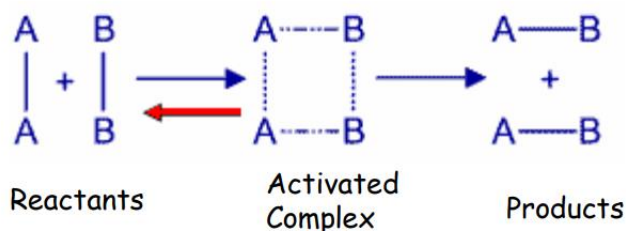


Activation Energy (E_a)

“The minimum amount of energy, in addition to the average kinetic energy, which the particles must possess for effective collisions, is called activation energy.”

Explanation:

- Chemical reactions involve the breaking and making of chemical bonds. These changes are accompanied by changes in energies.
- Collision theory has been proposed to explain the observed kinetics of reactions.
- For a chemical reaction to occur, the combining atoms or molecules must collide with one another. These collisions may be effective or ineffective depending upon the energy and orientation of the colliding particles.
- The effective collision can take place only if the energy of the colliding particles is high enough to overcome the repulsion between electrons around the reacting particles.
- Proper orientation means that at the time of collision, the atoms which are required to make new bonds should collide with each other.
- No reaction occurs if the energy of the reacting particles is lower than the activation energy. Thus, the speed of a reaction depends on its activation energy.
- The higher the activation energy, the lower the reaction rate.
- Consider the reaction between A_2 and B_2 molecules and a new molecule AB . If the energy of these molecules is equal to or greater than the activation energy, the collision break their bonds new bonds are formed.
- In an effective collision, the molecules form an unstable species called an activated complex.
- It is short lived and then converted into products.



Activation energy appears in the form of hills between reactants and products. Molecules must first climb over an energy barrier before they can roll down the hill to form products.

Only molecular collisions with the right activation energy and orientation can do this.

A potential energy diagram can also be used to understand enthalpy changes in chemical reactions.

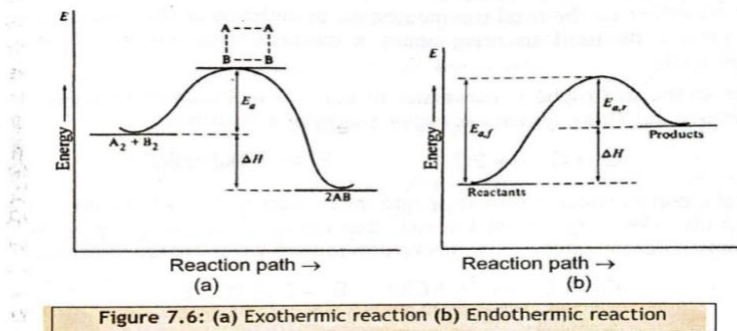
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The **reaction enthalpy** is equal to the difference between the energies of the reactants and the products.

In an **exothermic reaction**, the products are at a lower energy level than the reactants.

In an **endothermic process**, the products have a higher energy level than the reactants.

Endothermic reactions require a constant source of energy to complete the reaction.

**Catalysis**

“A substance which accelerates a chemical reaction but remains chemically unchanged at the end of a reaction is called as catalyst and the phenomenon is called catalysis.”

Explanation:

Another way to increase reaction rate and to lower the activation energy of the rate determining step is by introducing a catalyst.

It increases the rate of reaction by decreasing the energy of activation of a reaction.

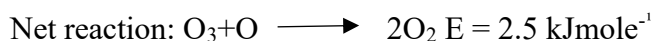
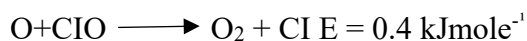
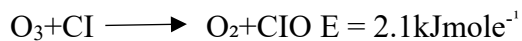
A catalyst has no effect on the total thermodynamics of chemical reaction. It cannot be used to bring about a chemical reaction, which is not favored thermodynamically.

For example, in the stratosphere, conversion of an ozone molecule by an oxygen atom into two O₂ molecules occurs. This reaction has higher energy of activation.



Chlorofluorocarbon compounds diffuse up into the stratosphere. These compounds absorb short wave length ultraviolet light from the sun, that breaks carbon-chlorine bonds and produce chlorine atoms.

Cl atom catalyze the mechanism requiring less energy of activation.



It shows that the direct reaction between O₃ and O has a substantially higher activation energy than the chlorine catalyzed reaction.

Gibbs free energy and feasibility of a reaction

- If ΔG° is negative or equal to zero, then the reaction is feasible.
- If ΔG° is positive, then the reaction is not feasible.

For More Information

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